

Swapnika Reddy Kolan

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Summary

Data Engineer with over 3 years of experience building data pipelines, ETL processes, and data lakes in cloud environments. Developed solutions in Python, SQL, and Spark to process large datasets. Leveraged AWS tools such as S3, Redshift, Glue, Lambda, and Athena to manage cloud data workflows. Built and scheduled workflows using Airflow and managed infrastructure with Terraform and CloudFormation. Designed Snowflake and Redshift data models using star and snowflake schemas. Implemented data validation with PyTest and Great Expectations. Collaborated with analytics teams to deliver datasets and reports in Power BI and Tableau. Followed Agile practices to support real-time reporting and analysis.

Technical Skills

- Programming & Scripting Languages:** Python (Pandas, NumPy, PySpark, Scikit-Learn), SQL (CTEs, Window Functions, Optimization), Scala, Java, Bash/Shell Scripting
- Big Data & Distributed Computing:** Databricks, Apache Spark (PySpark, Scala), Hadoop (HDFS, Hive, HBase, Impala), Kafka, Flink, Delta Lake
- Data Modeling & Databases:** PostgreSQL, MySQL, SQL Server, Oracle, MongoDB, DynamoDB, Cassandra, Snowflake, BigQuery, Amazon Redshift, Data Lakehouse
- ETL & Data Pipelines:** Apache Airflow, dbt, AWS Glue, Azure Data Factory, Talend, Informatica
- Cloud Technologies & Services:** AWS (S3, Redshift, Glue, Lambda, RDS, EMR, Athena), Azure (Synapse, Data Factory, Blob Storage, Cosmos DB), GCP (BigQuery, Dataflow, Cloud Storage)
- DevOps & CI/CD:** Docker, Kubernetes, Terraform, AWS CloudFormation, Jenkins, GitHub Actions, GitLab CI/CD
- Data Governance & Security:** Data Quality, Data Lineage, Metadata Management, RBAC, Compliance (GDPR, CCPA, HIPAA)
- Business Intelligence & Reporting:** Tableau, Power BI, Looker, Metabase
- Performance Optimization & Monitoring:** Query Tuning (Partitioning, Indexing, Caching), Prometheus, Grafana, ELK Stack, AWS CloudWatch

Experience

Liberty Mutual Finance

Jan 2024 - Present

Data Engineer

- Built ETL pipelines using AWS Glue, PySpark, Lambda, Python, and SQL to process over 10 TB of structured and semi-structured clinical data from 10+ sources, reducing manual data preparation by 80%.
- Designed a data lake on AWS S3, integrated with Redshift Spectrum and Athena, supporting 10+ analysts and enabling ad-hoc querying on datasets exceeding 15 TB.
- Developed dimensional models and curated datasets in Snowflake, cutting query times by 60% and supporting daily analytics for 5+ compliance and clinical teams.
- Engineered ETL workflows using AWS Glue, Apache Spark (PySpark/Scala), and Snowflake, enabling real-time dashboards in Power BI that surfaced trends across 120M+ patient records.
- Reduced data ingestion time from 4.5 to 1 hour (a 3.5-hour improvement) by tuning Spark jobs, adding Airflow orchestration, and automating CI/CD with Terraform, GitHub, and Jenkins.
- Delivered 12+ KPI dashboards in Power BI, supporting FDA compliance and internal audits; monitored pipeline uptime and performance using AWS CloudWatch, Prometheus, and Python-based alerts.
- Processed 20+ TB of historical healthcare data using Hadoop and Spark with Scala and Python, achieving up to 5x faster batch processing for downstream analytics.

Capri Global

Feb 2021 – Nov 2022

Data Engineer

- Engineered ETL pipelines using Apache Airflow, Python, and SQL to integrate transactional data from 1,000+ retail locations into Redshift.
- Developed scalable ETL workflows using Apache Airflow, SQL, and AWS Redshift to transform 60M+ daily retail transactions into Tableau-ready datasets for marketing, inventory, and sales analytics.
- Migrated legacy workflows to AWS S3 and Glue, improving scalability and reliability of daily data processing jobs.
- Built business-facing data marts and dimensional models for marketing, sales, and inventory analytics.
- Automated data quality validation using Python and Great Expectations, ensuring high-trust analytics delivery.
- Supported Tableau-based reporting by delivering clean, modeled data that improved campaign tracking and sales forecasting.

Education

Masters in Information Technology

Pittsburg State University

Bachelors in Computer Science and Engineering

Sphoorthy Engineering College

Certifications

- AWS Certified Data Engineer - Associate